QP CODE: 25900116

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC)–MG2DSCHPY100-PHYSIOLOGY OF HUMAN CONTROL SYSTEM

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Interest (I), Appreciation(Ap), and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple ChoiceQuestions
Answer all questions
Each question carries **1** mark

1. Option A - Stapedius
2. Option A – Somatic Nervous System
3. Option C - Pineal Gland
4. Option B – Afferent Nerve
5. Option A - Hormones
6. Option C - Mineralocorticoids

Part B
Short Answer Type Questions.
Answer any **2** questions.
Each question carries **2** marks.

7. Parageusia - Persistent abnormal taste in the mouth even in the absence of stimuli **(1 mark)**

   E.g: a continuous metallic or bitter taste **(1 mark)**

8. Rhombencephalon or hindbrain is subdivided into two portions:

   i. Metencephalon, formed by pons and cerebellum **(1 mark)**

ii. Myelencephalon or medulla oblongata. **(1 mark)**

9. Pancreas is both **exocrine**(secreting pancreatic juice containing digestive enzymes) and **endocrine** (producing several important hormones,including insulin,glucagon and somatostatin) **(2 marks)**

10. Parathormone is secreted by the Parathyroid gland
- Parathormone is the most important regulator of blood calcium levels
- The hormone is secreted in response to low blood calcium levels
- The hormone causes increase in calcium and phosphate levels of the blood
- The hormone promotes the absorption of calcium and other ions from the intestinal lumen **(1/2 mark for each point)**

Part C
Short Essay Type Questions.
Answer any **4** questions.
Each question carries **5** marks.

11. Parts of the Olfactory System

1) Nasal Epithelium - Also called Olfactory Epithelium.Specialized epithelial tissue inside the nasal cavity that is involved with the sense of smell.Part of the olfactory system directly responsible for detecting odours **(1 mark)**

2) Receptor cells- Olfactory receptor cells in the cell membranes of olfactory receptor neurons are responsible for the detection of odour molecules.Activated olfactory receptors are the initial player in a signal cascade,which ultimately produces a nerve impulse that is transmitted to the brain **(1 mark)**

3) Glomerulus - Spherical structure in the olfactory bulb.The glomerulus is the basic unit in the odour map of the olfactory bulb.Each odour activates a different pattern of glomeruli,which decode the identity of the odour.**(1 mark)**

4) Mitral Cells - The mitral cells are neurons that are part of the Olfactory system.Located in the olfactory bulb in the mammalian central nervous system.Receive information from the axons of olfactory receptor neurons ,forming synapses in neuropils called glomeruli.**(1 mark)**

5)Olfactory bulb- Present on the inferior side of the brain.Divided into two distinct structures,the main olfactory bulb and the accessory olfactory bulb.The Olfactory bulb transmits smell information from the nose to the brain.**(1 mark)**

12. Labelled diagram of neuron **– 1 mark**

- **Cell body** - known as soma or perikaryon
- Constituted by a mass of cytoplasm called neuroplasm
- The cytoplasm contains a large nucleus, Nissl bodies, neurofibrils, mitochondria and Golgi apparatus.
- Nissl bodies and neurofibrils are found only in nerve cell and not in other cells.
- **Dendrons -** branched process of neuron and it is branched repeatedly to form dendrites.
- Dendrites transmits impulses towards the nerve cell body
- **Axon -** the longer process of nerve cell.
- Axon arises from axon hillock of the nerve cell body
- Axon transmits impulses away from the nerve cell body.
- Axon has a long central core of cytoplasm called axoplasm.
- Axoplasm is covered by the tubular sheath like membrane called axolemma.
- Axon is divided into small branches called axon endings or telodendrons with bulb like structures at the end called synaptic knob or terminal buttons.
- Inside synaptic knobs, several secretory vesicles called synaptic vesicles are present. They secrete and store neurotransmitters like acetylcholine.
- Axon is covered by a membrane called neurilemma. It is also called neurilemmal sheath or sheath of Schwann. It contains Schwann cells.
- Some nerve fibres are covered by one more membrane called medullary or myelin sheath. They are called medullated or myelinated nerve fibres.
- Myelin sheath is a thick lipoprotein sheath that insulates the myelinated nerve fiber. It is absent at regular intervals.
- The area where myelin sheath is absent is called node of Ranvier. It acts as an insulating jacket for axon. **(1/4 mark for each point)**

13) Hormonal Disorders of Posterior Pituitary Gland

Hypoactivity of Posterior Pituitary Gland

Diabetes Insipidus

- Conditions in which there is a deficiency of the production of vasopressin

- Deficiency of vasopressin results in diuresis or polyuria and polydipsia(excessive drinking)

- Urine volume can exceed 15 L/day.

-Excretion of large amounts of dilute urine causes dehydration and increased concentration of sodium chloride in the extracellular fluid **(3 marks)**

Hyperactivity of Posterior Pituitary Gland

SIADH(Syndrome of Inappropriate Hypersecretion of Antidiuretic Hormone)

- Disease characterized by loss of sodium through urine due to hypersecretion of ADH.

- Signs and symptoms include loss of appetite,weight loss,nausea,vomitting,muscle weakness,restlessness and irritability.**(2 marks)**

14. Astigmatism

- The condition in which light rays are not brought to a sharp point upon retina. When it is moderate, it is known as physiological astigmatism. When it is well marked, it is considered abnormal.
- In a normal eye, lens has approximately same curvature in all meridians. So, the light rays are refracted almost equally in all meridians and brought to a focus.
- If the curvature is different in different meridians, vertical, horizontal and oblique, the refractive power is also different in different meridians.
- The meridian with greater curvature refracts the light rays more strongly than the other meridians. So, these light rays are brought to a focus in front of the light rays, which pass through other meridians. Such irregularity of curvature of lens causes astigmatism.
- Astigmatism is corrected by using cylindrical glass lens having the convexity in the meridians, corresponding to that of lens of eye having a lesser curvature, i.e. if the

horizontal curvature of lens is less, the person should use cylindrical glass lens with the convexity in horizontal meridian. **(1 mark for each point)**

15. Synapse - The junction between two neurons is called synapse. It is through this junction that an impulse pass from one neuron to other. (**1 Mark)**

- Classification of Synapse - Anatomical classification and Functional classification **(1/2 mark)**

Anatomical Classification

a) Axoaxonic synapse in which axon of one neuron terminates on axon of another neuron. **(1/2 mark)**

b) Axodendritic synapse in which the axon of one neuron terminates on dendrite of another neuron. **(1/2 mark)**

c) Axosomatic synapse in which axon of one neuron ends on soma (cell body) of another neuron. **(1/2 mark)**

Functional classification

a) Electrical synapse

- Synapse in which the physiological continuity between the presynaptic and the post synaptic neurons is provided by gap junction between the two neurons.
- Direct exchange of ions between the two neurons through the gap junction.
- The action potential reaching the terminal portion of presynaptic neuron directly enters the postsynaptic neuron. Important feature of electrical synapse is that the synaptic delay is very less because of the direct flow of current.
- This type of impulse transmission occurs in some tissues like the cardiac muscle fibers, smooth muscle fibers of intestine and the epithelial cells of lens in the eye. **(1 mark – ¼ mark for each point)**

b) Chemical synapse

- junction between a nerve fiber and a muscle fiber or between two nerve fibers through which the signals are transmitted by the release of chemical transmitter.
- No continuity between the two neurons because of the presence of a space called synaptic cleft between the two neurons.

- Action potential reaching the presynaptic terminal causes release of neurotransmitter substance
- Neurotransmitter reaches the postsynaptic neuron through synaptic cleft and causes the production of potential change. **(1 mark – ¼ mark for each point)**

16) Hormonal disorders of Thyroid gland

1) Goitre - An enlargement of the thyroid gland is called goitre. If there is an iodine deficiency,the thyroid cannot make sufficient hormone.This stimulates the anterior pituitary to secrete thyroid stimulating hormone,which causes the thyroid gland to increase in size to produce more hormones.But it cannot produce more hormones because it doesnot have the necessary raw material,iodine.This type of thyroid enlargement is called simple goitre or iodine deficiency goitre. **(1 mark)**

2) Hyperthyroidism(Exopthalmic Goitre)-It is characterised by an enlargement of the entire thyroid gland.The overactive thyroid releases too much T3 and T4 into the blood stream.It results in the acceleration of metabolic rate.Also called Graves Disease.There is swelling of tissues in the orbits,frequent bulging of eye balls,weight loss,nervousness and rapid pulse. **(1 mark)**

3) Hypothyroidism (Cretinism) - An underactive thyroid releases too little T4 and T3 into the blood stream.This condition is called Hypothyroidism.This causes the metabolism to slow down too much Hypothyroidism at birth results in **cretinism** characterised by retarded growth,severe nental retardation and delayed sexual maturity**.(1 mark)**

In adults,hypothroidism results in a condition known as **myxoedema,**characterised by physical sluggishness,low metabolic rate,mental dullness,dry and coarse skin and puffy face **(1 mark)**

4) Thyroid Adenoma - Sometimes ,a localized tumor develops in the thyroid tissue.It is known as Thyroid adenoma and it secretes large quantities of thyroid hormones.It is not associated with autoimmunity.As far as this adenoma remains active,the other parts of the thyroid gland cannot secrete the hormone**.(1 mark)**

Part D

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Functions of Skin

1. **Protective Function**
   - **a)** Protection from Bacteria and Toxic Substances
   - **b)** Protection from Mechanical Blow
   - **c)** Protection from Ultraviolet Rays **(2 marks)**
2. **Sensory Function** - Skin has many nerve endings, which form the specialized cutaneous receptors. These receptors are stimulated by sensations of touch, pain, pressure or temperature sensation and convey these sensations to the brain via afferent nerves.
3. **Storage Function** - Skin stores fat, water, chloride and sugar. It can also store blood by the dilatation of the cutaneous blood vessels.
4. **Synthetic Function** - Vitamin D3 is synthesized in skin by the action of ultraviolet rays from sunlight on cholesterol. Exposure to the sun ensures supply of vitamin D.
5. **Regulation of Body Temperature** - Excess heat is lost from the body through skin by radiation, conduction, convection and evaporation. Cutaneous vasoconstriction diverts the blood to the interior of the body and so diminishes heat loss. Vasodilatation of the skin helps in elimination of heat from the body. Sweat glands of the skin play an active part in heat loss, by secreting sweat. Hairs of skin regulate body temperature by entangling layer of air
6. **Regulation of Water and Electrolyte Balance** - Skin regulates water balance and electrolyte balance by excreting water and salts through sweat
7. **Excretory Function** - Skin excretes small quantities of waste materials like urea, salts and fatty substance through sweat and insensible perspiration.
8. **Absorptive Function** - Lipids are easily permeable through the skin. Lipid-soluble substances like vitamins are easily absorbed through the skin
9. **Secretory Function** - Skin secretes sweat through sweat glands and sebum through sebaceous gland, The sebaceous glands secrete a fatty substance which is rich in cholesterol called sebum. It helps to keep the skin moist, smooth and prevents drying. In cold climates the secretion is depressed and the skin becomes dry and scaly. **(1 mark each for point 2 -9)**

18. Neurotransmitter is a chemical substance that acts as a mediator for the transmission of nerve impulse from one neuron to another neuron through a synapse. **(1 mark)**

Classification

1. **Depending upon Chemical Nature**
   i) Amino Acids
   ii) Amines
   iii) Others **(3 marks)**
2. **Depending upon Function**

i) Excitatory Neurotransmitters

ii) Inhibitory Neurotransmitters **(2 marks)**

Important Neurotransmitters

1. Acetylcholine
2. Noradrenaline
3. Dopamine
4. Serotonin
5. Histamine
6. Gamma-aminobutyric Acid
7. Nitric Oxide **(any four – 1 mark each)**

19. Functions of Ovaries

1. Production of Eggs(Oogenesis)- The ovaries produce and release eggs during the menstrual cycle **(2 mark)**

2. **Hormone Production -**

- **Estrogen-**responsible for the development of female secondary sexual characteristics,promotes the growth of endometrium in preparation for pregnancy

-**Progesterone-**prepares the endometrium for the implantation of a fertilized egg and helps maintain early pregnancy,helps regulate the menstrual cycle and prevents further ovulation during pregnancy

-**Inhibin -** Helps regulate the levels of FSH,inhibiting excessive egg production and maintaining hormonal balance

-**Relaxin-**produced during pregnancy to relax the uterine muscles and the ligaments of the pelvis,facilitating child birth **(2 mark)**

3. Regulation of the Menstrual Cycle- The ovaries work in conjuction with hypothalamus and pituitary gland to regulate the menstrual cycle.Hormones like estrogen and Progesterone are involved in the cyclical changes in the endometrium**(2 mark)**

4. Support of Pregnancy(if fertilization occurs)- The ovaries,continue to produce progesterone to maintain the uterine lining and support the early stages of pregnancy.If pregnancy occurs,the ovaries continue to secrete hormones(including progesterone and estrogen)until the placenta takes over hormone production**(2 mark)**

5. **Release of hormones that impact overall health-** Estrogen and Progesterone not only regulate the menstrual cycle,but also play a role in bone health,and maintaining the health of the skin,mood and energy levels.

The balance of these hormones is critical in preventing conditions such as osteoporosis,heart disease and other metabolic disorders.**(2 mark)**

20. Transmission of nerve impulse

Impulse transmission in the nervous system involves the process by which electrical signals or nerve impulses are transferred between neurons or from neurons to muscles or glands. This is a complex process that relies on both electrical and chemical signaling mechanisms. **(1 mark)**

The steps involved are:

1. Resting Membrane Potential
2. Action Potential Initiation
3. Propagation of the Action Potential
4. Synaptic Transmission
5. Postsynaptic Response
6. Reuptake and Degradation of Neurotransmitters **(1 ½ mark for each point)**